

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2022

ESE 4219

(Energy Audit and Management)

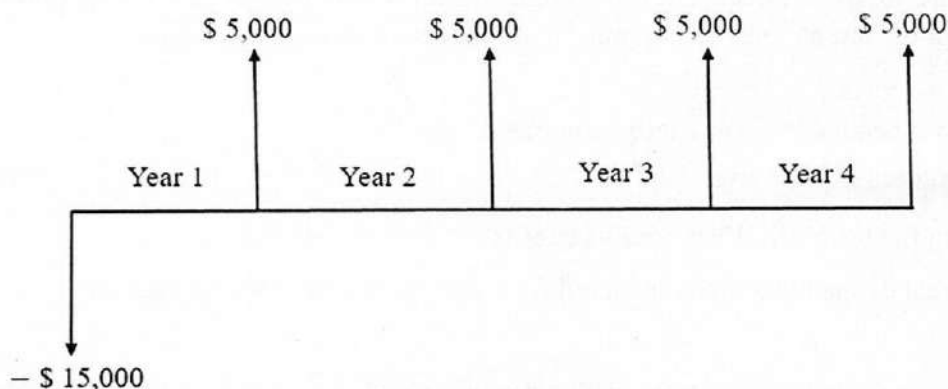
Time: 3 Hours.

Full Marks: 210

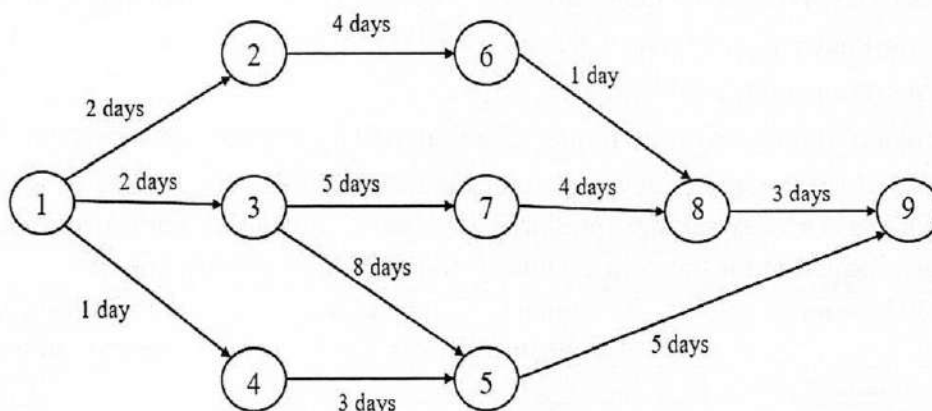
- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is meant by energy management? What are its objectives? 08
1(b). What information should be collected during a detailed energy audit? 10
1(c). Write down the functions of following energy audit instruments: 10
(i) Combustion analyzer, (ii) Lux meter, (iii) Conductivity meter, and (iv) Thermal insulation scanner.
1(d). How energy conservation opportunities can be identified in energy audit? 07
- 2(a). What is meant by external benchmarking? What factors should be considered in external benchmarking? 07
2(b). What is meant by ROI? What are its advantages and limitations? 10
2(c). With proper example explain the time value of money? 08
2(d). Calculate the IRR of a project from the following cash flow diagram. 10



- 3(a). What are the functions of ESCO? 09
3(b). What is PERT? What are the benefits and limitations of PERT? 10
3(c). Define the following terms related to project management: 06
(i) Slack, (ii) Critical path and (iii) Project crashing.
3(d). Find the critical path and calculate the slack time for the following network. 10



- 4(a). Why is reactive power control necessary in power system? 08
4(b). Explain the various techniques to improve power factor. 07

- 4(c). An oil fired reheating furnace heats steel billets from 40°C to 1220°C at a furnace efficiency of 28%. The furnace operate for 4700 hours per annum. The GCV of furnace oil is 10,000 kCal/kg and density is 0.94 g/litre. The cost of furnace oil is BDT 45/litre. The specific heat of billet is 0.12 kCal/kg $^{\circ}\text{C}$. 20
- (i) Calculate the amount of energy needed to heat 12 tons of steel billets per hour.
- (ii) Calculate litres of furnace oil fired per tons of steel billets.
- (iii) If the efficiency of furnace is improved from 28% to 30% by adopting ceramic fibre insulation, calculate hourly furnace oil cost saving.
- (iv) what is the simple payback period if the investement is BDT 20 lakhs.
- (v) How large could be the investment to improve the efficiency at an internal rate of 16% and per year over 6 years.

SECTION – B

- 5(a). What is the purpose of material and energy balance? Distinguish between continuous process and batch process. 08
- 5(b). How can energy be lost in a system? 06
- 5(c). Briefly explain why appointment of energy manager is needed for successful energy management program. 05
- 5(d). Explain the three key factors that motivate the employees. 06
- 5(e). A textile dryer is found to consume $5\text{m}^3/\text{hr}$ of natural gas with a calorific value of 900 kJ/mole. If the throughput of the dryer is 70 kg of wet cloth per hour, drying it from 60% moisture to 10% moisture, estimate the overall thermal efficiency of the dryer taking into account the latent heat of evaporation only (Assume reasonable value). 10
- 6(a). Why is it needed to form energy committee? 05
- 6(b). With a force field analysis chart briefly explain how a force field analysis is carried out? 15
- 6(c). Explain the term 5S. What are its benefits? 10
- 6(d). Why heat exchangers are considered for waste heat recovery application? 05
- 7(a). What is meant by economic thickness of insulation? What factors influence economic thickness of insulation? 08
- 7(b). What modifications are proposed in BNBC for building energy efficiency? What is EPI? Explain. 10
- 7(c). Briefly explain the main attributes of green building? How does green construction reduces the use of energy and pure material? 12
- 7(d). Is it possible to consider a pressure cooker as a boiler? Explain your answer. 05
- 8(a). What is COP? What is the major difference between convention and protocol? 08
- 8(b). Write short notes on: (i) UNFCCC, and (ii) CDM 06
- 8(c). Write short note on the followings: 12
- (i) Emission trading, (ii) Joint implementation, (iii) Sustainable development.
- 8(d). Use CUSUM technique to develop a table and calculate energy savings for first 6 months of 2023 for those energy savings measures implemented by a plant prior to January 2023. The average production for the period January-June 2023 is 1000 MT/month. 09

2023- month	Actual specific energy consumption kWh/MT	Predicted specific energy consumption kWh/MT
January	1203	1121
February	1187	1278
March	1450	1550
April	1401	1571
May	1350	1284
June	1240	1233

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2022

ESE 4003

(Hydrogen and Fuel Cells)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|------|--|----|
| 1(a) | Explain hydrogen production challenges. Discuss the benefits and barriers of hydrogen fuel usage. | 09 |
| 1(b) | Explain hydrogen production by steam methane reforming. | 16 |
| 1(c) | Discuss the significance of feedstock desulfurization and the role of catalytic reforming in steam methane reforming. | 10 |
| 2(a) | What is meant by biomass gasification? Explain hydrogen production by biomass gasification. | 12 |
| 2(b) | How does hydrogen purification via pressure swing adsorption compare with membrane separation in terms of efficiency? | 15 |
| 2(c) | What are the advantages and disadvantages of TiO ₂ as photo catalyst for water splitting process? | 08 |
| 3(a) | What are the different methods of hydrogen storage, and how do they vary in terms of efficiency, safety and applicability? | 12 |
| 3(b) | Briefly explain the principles behind compressed hydrogen storage and how it compare to other storage methods? | 10 |
| 3(c) | What are the main applications of hydrogen in various industries, including energy, transportation, and manufacturing? | 13 |
| 4(a) | Describe the pressure swing adsorption (PSA) method for hydrogen purification. | 10 |
| 4(b) | Discuss the desired characteristics of sorbents for CO ₂ absorption system. Explain hydrogen storage system using metal hydrides. | 14 |
| 4(c) | Discuss the challenges associated with different hydrogen transportation systems. | 11 |

SECTION – B

- | | | |
|------|---|----|
| 5(a) | What fundamental thermodynamic principles govern the operation of fuel cell? | 06 |
| 5(b) | Derive an expression for maximum possible thermodynamic efficiency of a fuel cell. | 10 |
| 5(c) | What is fuel cell subsystem? Why the gross fuel cell stack electrical efficiency differs from the net fuel cell subsystem electrical efficiency? Explain. | 09 |

5(d)	A direct methanol fuel cell uses methanol as fuel where temperature is 30°C and pressure is 100 kPa. Calculate half-cell potential and compare anode, cathode and total thermodynamic potential for both medium. ΔG_f Values: $\text{CH}_3\text{OH} = -174 \text{ kJ/mole}$, $\text{CO}_2 = -394 \text{ kJ/mole}$, $\text{H}_2\text{O} = -237 \text{ kJ/mole}$, $\text{OH}^- = -157 \text{ kJ/mole}$. From thermodynamic point of view, tell which reaction is more favorable and why?	10
6(a)	Explain the construction of a proton exchange membrane fuel cell. Briefly discuss its working principle.	14
6(b)	What are the two main goals of fuel cell characterization? List and explain at least three major operation variables that can affect fuel cell performance.	12
6(c)	Discuss the relative advantages and disadvantages of EIS versus current interrupt measurement.	09
7(a)	Discuss the factors influencing the voltage output of a fuel cell. Explain how cell voltage varies with changing operating condition.	10
7(b)	Derive the Butler-Volmer equation within the context of fuel cells and explain how this equation elucidates the kinetics of electrode reactions between current density and over potential.	15
7(c)	Consider a hydrogen fuel cell with an exchange current density (i_0) of $1 \times 10^{-5} \text{ A/cm}^2$ at room temperature (298K). Calculate the over potential (η) required to achieve a current density (i) of 0.001 A/cm^2 for the hydrogen oxidation reaction. [$\alpha = 0.5$].	10
8(a)	What is cogeneration? Explain the design of a combined heat and power (CHP) fuel cell system with necessary diagram.	12
8(b)	Explain the integrated gasification carbonate fuel cell concept.	11
8(c)	How do hybrid gas turbine fuel cell (GTFC) systems contribute to reducing emissions and environment impact compared to environmental power generation method?	12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2022

ESE 4013

(Energy in Built Environment)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

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|------|---|----|
| 1(a) | What is meant by 'Indoor Environment control'? To control indoor environment, provide some technical suggestions with examples. | 08 |
| 1(b) | What is meant by thermal comfort? Explain the impact of thermal control on building design. | 08 |
| 1(c) | Discuss the aspects of buildings design on the followings:
(i) thermal comfort, (ii) visual comfort, (iii) acoustic comfort. | 09 |
| 1(d) | Based on indoor activities, categories the building and mention the energy saving options for each category. | 10 |
| 2(a) | For an existing building, discuss the technical issues to reduce energy consumption. Explain the strategies to ensure good acoustic environment in building. | 07 |
| 2(b) | What is meant by acoustic comfort? Discuss about different building materials that can be used to maintain good acoustic comfort. | 08 |
| 2(c) | Graphically show and discuss the variation of temperature and heat flow of a wall for a typical day from mid-night to mid-night. | 08 |
| 2(d) | Discuss the design considerations of a building wall and roof for different climate conditions. Also discuss the effect of wall and roof materials on visual comfort. | 12 |
| 3(a) | On the basis of energy consumption view point, signify the sun-earth relationship for a building. | 10 |
| 3(b) | What are meant by horizontal shadow angle and vertical shadow angle? How to calculate them and mention the significance of them. | 10 |
| 3(c) | How to calculate sky temperature? How sky temperature influences the analysis of solar energy utilization in building? | 08 |
| 3(d) | Discuss about the factors that should be considered to design and select a sun-shading device. | 07 |
| 4(a) | How to identify the size of shading devices? Discuss with the help of sketchmatic diagram | 10 |
| 4(b) | With the help of neat sketch, describe the active daylighting system. | 10 |
| 4(c) | What factors should be considered while designing a natural daylighting system? Explain with proper example. | 08 |

- 4(d) How to increase natural wind flow around building? Discuss with schematically. 07

SECTION – B

- 5(a) How to calculate heat losses from the vertical basement walls? 09
- 5(b) What is the importance of internal heat gains in building heating and cooling load calculation? What are the main components of internal heat gains? 09
- 5(c) How color and number of glazing affect the heat transfer through window? Write short note on “smart window”. 10
- 5(d) How to calculate sensible heat gain and latent heat gain in a building? 07
- 6(a) With the help of schematic diagram, mathematically explain the thermal transient heat transfer through wall. 09
- 6(b) What is meant by solar heat gain coefficient (SHGC)? How to calculate SHGC of a single pane glass window? 09
- 6(c) What factor influences the air infiltration through building envelope? Explain the tracer gas method to measure the infiltration rate. 09
- 6(d) What does mean by zoning of building? Why it is required? Explain with example. 08
- 7(a) What is stack effect? Explain with schematic diagram. 06
- 7(b) What is balance point temperature? Explain the degree-day method to estimate the heating and cooling load of building. Differentiate between degree-day method and bin method. 13
- 7(c) Distinguish between infiltration and natural ventilation. Explain air change method for calculating air infiltration rate. 12
- 7(d) What are the desirable characteristics of window glasses in cold location? 04
- 8(a) Why net-zero energy this building important? What energy efficiency measures should be taken in the building? How to design a green building? 12
- 8(b) Explain the evaporative cooling system with solar-assisted desiccant dehumidification system. 08
- 8(c) What are the functions of LEED? How to achieve a LEED certificate? Discuss. 07
- 8(d) How to overcome the economic challenges of building by building energy management? 08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2023

ESE 4213

(Nuclear Power Engineering)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|---|----|
| 1(a). Write short note on the following terms:
(i) Fission reaction, (ii) Fusion reaction | 04 |
| 1(b). What is nuclear strong force? Write down its properties. | 07 |
| 1(c). Why do some nuclei undergo fission reaction while others undergo fusion reaction? Describe with the help of necessary figure. | 12 |
| 1(d). Elaborate on the neutron life cycle where $k=1$. | 12 |
| 2(a). Write short note on the following terms:
(i) Radioactivity, (ii) Absorbed dose and (iii) Effective dose | 06 |
| 2(b). Briefly explain the factors affecting external dose uptake. | 06 |
| 2(c). What is fundamental safety objective? Describe the safety principle “Responsibility for safety”. | 11 |
| 2(d). Describe the biological effect of ionizing radiation. | 12 |
| 3(a). What is uranium ore? Give some examples of major primary ore mineral of uranium. | 03 |
| 3(b). Why is fuel enrichment required? With neat sketch, describe the gas diffusion method of fuel enrichment. | 11 |
| 3(c). Describe a PWR nuclear fuel assembly with necessary figure. | 10 |
| 3(d). With the help of a flow diagram, explain the reaction in standard UO_2 fuel. | 11 |
| 4(a). What can be done with used nuclear fuel? | 04 |
| 4(b). What is MOX fuel? Describe reaction in MOX fuel with appropriate flow diagram. | 12 |
| 4(c). What is radioactive waste? Classify it. | 09 |
| 4(d). Why multi-purpose canisters are used? Describe the deep geological disposal system. | 10 |

SECTION – B

- | | | |
|-------|---|----|
| 5(a). | Classify different types of nuclear reactor according to the progression of time. | 06 |
| 5(b). | Which type of nuclear reactor is mostly used throughout the world? Discuss the reasons behind it. | 08 |
| 5(c). | What are the different components used in nuclear reactor? Discuss a comparative analysis between Uranium and Plutonium fuel in terms of critical mass analysis. | 11 |
| 5(d). | What are the different heat transfer enhancement techniques used in nuclear reactor? Discuss it. | 10 |
| | | |
| 6(a). | What are the different features of Generation III technologies that are different from Generation II technologies? | 07 |
| 6(b). | What are the challenges and opportunities in implementing passive safety systems in nuclear reactors, and how can these system be further optimized to eliminate the reliance on active intervention during emergencies? | 10 |
| 6(c). | Describe the gas-cooled fast reactor with proper diagram. | 10 |
| 6(d). | Discuss the different features of European pressurized water reactor (EPR). | 08 |
| | | |
| 7(a). | How can Generation IV nuclear reactor designs such as molten salt reactors or high temperature gas-cooled reactors address the challenges of waste management and fuel efficiency while ensuring enhancement safety compared to conventional reactor types? | 12 |
| 7(b). | What will happen in a nuclear power plant if reactor coolant pump does not properly work? | 10 |
| 7(c). | Describe the reactor vessel internal with their functions. | 13 |
| | | |
| 8(a). | In which cane burnable poison formed? How can it be controlled? | 08 |
| 8(b). | Briefly discuss about Reactor Control and Compensation System (RCCA) in nuclear reactor. | 12 |
| 8(c). | Write down the differences between PWR and BWR. Which one is better in the context of environmental safety? Describe. | 15 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2022

ESE 4017

(Energy Efficiency Assessment)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Why it is necessary to study the energy performance assessment of a system? Explain with examples? 07
- 1(b). What is the significance of the evaporation ratio in assessing boiler performance, and how does a drop in evaporation ratio indicate a decrease in boiler efficiency? 08
- 1(c). The following are the data collected for a boiler using furnace oil as fuel ultimate analysis of fuel: 20
- Carbon : 84% ,
Hydrogen : 12%,
Nitrogen : 0.5%
Oxygen : 1.5%,
Sulphur: 1.5 %.
Moisture : 0.5%
CV of fuel : 9700 kcal/kg
Humidity : 0.025 kg moisture /kg of dry air
Flue gas analysis :
CO₂ : 9.8% volume
Flue gas temperature : 195°C
Ambient temperature : 32°C
- (i) Determine the boiler efficiency based on GCV by indirect method.
(ii) If feed water temperature is 90°C and steam generation rate of 65 TPH at boiler pressure 60 kg/cm² and steam temperature 500 °C, what is the hourly consumption of furnace oil? The heat of steam at 60 kg/cm² and 500°C is 817 kcal/kg.
- 2(a). Differentiate between direct and indirect methods of performance evaluation for a boiler. Provide a example calculation for each method showing the % of accuracy. 07
- 2(b). How does the pressure of silica in boiler water contribute to the formation of hard silica scales, and what problems can it came? 09
- 2(c). Describe the effects of dissolved oxygen and carbon dioxide in boiler water including the corrosion mechanisms associated with them. 09
- 2(d). Describe the working priciples of a tray-type deaerator and how it heps in removing dissolved gases from boiler feed water. 10
- 3(a). Briefly discuss about different typs of membraines used in reverse osmosis process of water treatment. 10
- 3(b). Analyze the potential benefits and drawbacks of ceramic recuperator in comparison to metallic ones. Under what conditions and temeperature ranges do ceramic recuperator outperform their metallic counterpaths. 10

3(c).	Explain the followings to increase the boiler efficiency: (i) Excess air control (ii) Reduction of boiler steam pressure (iii) Speed control for fans, blowers and pump (iv) Boiler loading (v) Stack temperature	15
4(a).	Explain the significance of pipeline sizing in steam distribution system. How does oversized pipework differ from undersized pipework in terms of its impact on the system.	10
4(b).	For a steam distribution system, write short notes on the followings : (i) Drain points, (ii) Strainers, (iii) Branch lines, (iv) Separators.	08
4(c).	How do the fan's first, second and third law's describe the relationship between the fan performance, speed pressure and power?	12
4(d).	Write down the energy saving opportunities of a fan by (i) damper flow control, (ii) inlet guide vanes flow control.	05

SECTION – B

5(a).	Discuss about the characteristics of a energy efficient pump. How the different flow control strategies contribute on energy saving opportunities of a pump?	12
5(b).	What are the different flow measurement process? Explain briefly.	13
5(c).	With the help of necessary diagram, discuss about pump curve and system characteristics curve. How to select the operation point from these curves?	10
6(a).	How the performance of a refrigerant plant is assessed? Describe with required equation.	10
6(b).	What are the different factors affect performance and efficiency of a refrigeration plant? Briefly discuss any these of these factors.	13
6(c).	Discuss the different energy saving opportunities in cold storage system.	07
6(d).	Define Ice bank and explain how it works.	05
7(a).	List the basic parameters required to access the refrigeration capacity of achieved water system? Also, discuss them in briefly.	10
7(b).	How is the performance of power factor capacitor assessed? Explain	08
7(c).	Briefly discuss the factor affect the energy efficiency and minimize losses in operation of electrical motors.	12
7(d).	What are the different criteria that should be considered before selecting a motor.	05
8(a).	Define over motoring. Explain different types of motor speed control systems.	12
8(b).	How to select the capacity of diesel generator? Discuss the basic features of an energy efficient diesel generator.	08
8(c).	How does efficiency loss occur in a motor due to unbalance voltage? How to reduce this loss? Explain.	10
8(d).	Mention different star labelling of energy efficient indication motors.	05

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